

# LD9-V0 Mesh Fidelity and Topology Validation Specification

Sampled surface distance, normal error, scalar residual, and indexed-mesh topology

mdstats development specification

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## 1 LD9-V0 - Mesh Fidelity and Topology Metrics

**Package target:** `mdstats 0.19.54a0`

**Status:** implemented calibration metrics; simplification is not implemented here

**Module:** `mdstats.plotting.density_mesh_validation`

### 1.1 Purpose

This module defines deterministic validation metrics for comparing a candidate browser mesh with a raw reference mesh or, when supplied, the underlying scientific scalar field.

The module does not modify either mesh and does not perform mesh simplification.

### 1.2 Input constraints

Vertices are finite arrays with shape `(n_vertices, 3)`. Faces are integer arrays with shape `(n_faces, 3)`. Every face index must be valid, and a face may not repeat a vertex index. Zero-area triangles are rejected for geometric comparison.

### 1.3 Topology summary

```
MeshTopologySummary(  
    vertex_count,  
    edge_count,  
    face_count,  
    connected_component_count,  
    boundary_edge_count,  
    nonmanifold_edge_count,
```

```
euler_characteristic,
)
```

Undirected edge incidence is counted from indexed triangles. The Euler characteristic is

$$\chi = V - E + F.$$

A mesh is reported as a closed two-manifold when every used edge has incidence two.

This summary is a validation diagnostic. Periodic seam pairing remains a separate LD9-V1/V2 requirement.

## 1.4 Surface sampling

Triangles are sampled proportionally to area. For a selected triangle with vertices (**a**, **b**, **c**), two deterministic pseudorandom variates are reflected into the unit simplex and the point is

$$\mathbf{x} = \mathbf{a} + u(\mathbf{b} - \mathbf{a}) + v(\mathbf{c} - \mathbf{a}).$$

The seed and maximum sample count are explicit. Identical meshes sampled with the same seed produce identical point sets.

## 1.5 Symmetric sampled distance

Let  $P$  and  $Q$  be the sampled reference and candidate point sets. The sampled symmetric distance collection is

$$D(P, Q) = \{\min_{q \in Q} \|p - q\| : p \in P\} \cup \{\min_{p \in P} \|q - p\| : q \in Q\}.$$

The report stores median, 99th percentile, and maximum values. This is a bounded calibration approximation to symmetric surface distance; later production validation may add adaptive refinement near high-error regions.

## 1.6 Normal error

Triangle normals accompany every sampled point. Each sample is paired with its nearest opposite-mesh sample. Orientation-independent angular error is

$$\theta = \cos^{-1} (|\mathbf{n}_1 \cdot \mathbf{n}_2|).$$

The report stores median, 99th percentile, and maximum angles in degrees.

## 1.7 Scientific scalar residual

When a scalar sampler and contour level  $\rho_q$  are supplied, candidate samples are checked using

$$r(\mathbf{x}) = |\rho(\mathbf{x}) - \rho_q|.$$

The sampler must return one finite value per candidate sample. Scalar residual validation is optional in V0 but becomes required for LD9-V2 acceptance against the scientific field.

## 1.8 Fidelity policy

```
MeshFidelityOptions(  
    max_samples=50_000,  
    random_seed=0,  
    max_surface_error=0.02,  
    max_normal_error_degrees=8.0,  
    max_scalar_residual=None,  
    require_component_count=True,  
    require_euler_characteristic=True,  
    require_closed_two_manifold_match=True,  
)
```

The report is passing only when every enabled requirement passes.

## 1.9 Schemas

```
mdstats.mesh-topology-summary.v1  
mdstats.mesh-fidelity-options.v1  
mdstats.mesh-fidelity-report.v1
```

## 1.10 Focused tests

Tests must include:

- a closed octahedron with  $V = 6$ ,  $E = 12$ ,  $F = 8$ , and  $\chi = 2$ ;
- identical-mesh zero-distance comparison;
- translated-mesh surface-error rejection with preserved topology;
- scalar-residual evaluation against an explicit contour function;
- malformed face and zero-area rejection;
- canonical JSON round trips for topology, options, and fidelity reports.

## 1.11 Limitations and next stage

The V0 nearest-sample metric is suitable for calibration and regression tests but is not a formal exact Hausdorff computation. LD9-V2 should supplement it with adaptive field-based sampling, periodic seam checks, and protected-component validation.